Rewarding Learning

General Certificate of Secondary Education
2018

Digital Technology

Unit 1:

(Compulsory Core)

Digital Technology

[CODE]

SPECIMEN

MARK SCHEME

		AVAILABLE MARKS
1	**(a)** B The Copyright, Designs and Patents Act. [1]	
	(b) A It has no moving parts and consumes very little power. [1]	
	(c) D It carries data in pulses of light and is not prone to interference. [1]	
	(d) A Controls the way in which personal data is processed by the organisation. [1]	
	(e) D Enforces the Data Protection Act. [1]	5

2

STATEMENT	TICK (✓)
Information is raw facts and figures that have not been given a meaning.	
Information can be produced by processing raw data.	✓
Information is automatically encrypted when it is produced.	
Information is raw facts and figures that have been given a context.	✓

[2] 2

3 **(a)** Can be edited at pixel level [1]
Can adjust the resolution of picture being taken [1] [2]

(b) Any **two** from:

- Vector graphics do not depend on resolution for quality [1]
- Vector graphics store graphics in terms of objects [1]
- Objects can be easily moved around using vector graphics [1]
- Vector graphics can be easily resized without impacting on quality [1]

[2] 4

4 Mobile phones have a built-in **microphone** [1] to record sound.

Sound is input in **analogue** [1] format and is then converted into

digital [1] format so it can be stored in the phone's memory. [3] 3

			AVAILABLE MARKS
5	Any **two** from: • Provide an interface between the user and the computer [1] • Allocate internal memory (RAM) to programs and data currently being used [1] • Controls peripheral devices [1] • Error handling [1] • Management of security [1]	[2]	2
6	**(a)** 7	[1]	
	(b) Any from: Genre [1] = children [1] or Price [1] <= £2.99 [1] or Date published [1] between May and August 2012 [1]	[2]	
	(c)		

	Tick two boxes only(✓)
Range Check	✓ [1]
Heading Check	
Sort Check	
Report Check	
Presence Check	✓ [1]

		[2]	5
7	**(a)** Centre/centre across selection/merge and centre	[1]	
	(b) =sum [1] (F2:F7) [1] or =sum [1] (F2+F3+F4+F5+F6+F7) [1]	[2]	
	(c) The formula has been copied and pasted into the other cells in the column/the formula has been dragged down	[1]	
	(d) It is an absolute cell reference/the cell reference will not change [1] when the formula is copied/filled down into the other cells [1]	[2]	
	(e) A2:A6 [1] F2:F6 [1]	[2]	
	(f) The gardener could change the unit price/number needed [1] to see how this might change the total cost of the materials [1]	[2]	10

AVAILABLE MARKS

8 **(a)** Central Processing Unit [1]

(b) Every piece of information that is stored or processed goes through the CPU/Carries out the instructions of a computer's programs [1]

(c)

Statement	Part of the CPU that carries out this role
Holds programs and data currently in use	Immediate Access Store [1]
Performs calculations and logical operations	ALU [1]
Tells input and output devices how to carry out instructions	Control Unit [1]

[3]

(d) When a computer program is running the **Program Counter** [1] holds the address of the next instruction. The location of the next item of data to be fetched is stored in the **Memory Address Register** [1]. When a data item is fetched from memory it is stored in the **Memory Data Register** [1]. [3]

(e) ROM is non-volatile/the boot up program cannot be altered in ROM [1]

(f) **(i)** Storing most recently used instructions/storing most frequently used instructions [1]

(ii) Cache memory provides high speed access to instructions/ instruction fetches from cache are faster than instruction fetches from main memory [1]

Increased cache size means more instructions can be stored for faster access by the processor [1] [2] 12

9 **(a)** Any **two** from:

- Concept where all physical objects will be connected to the internet [1]
- Devices will be able to identify themselves to other devices [1]
- Devices use unique identifiers to communicate or transfer data [1] [2]

AVAILABLE MARKS

(b) (i) Any **one** from:

- Paul is not using his data allowance
- More reliable signal
- Greater download speed so better for streaming
- Download speeds not affected by mobile phone network traffic [1]

(ii) Any **two** from:

- Wi-fi range may be limited
- Signal strength may vary depending on proximity to router
- Could be subject to interference from other wireless sources
- Signal may be affected if there are a large number of users [2]

(c) (i) Any **two** from:

- Single cable called a trunk or backbone is used to connect all nodes
- Signals travel along the backbone in both directions
- Terminators at end of the backbone
- As signals/messages are passed along the backbone each node will examine the message/signal
- Node will only accept the message if it is addressed to it/if message addressed to another node it will be ignored
- If the backbone fails it will affect the entire network [2]

(ii) Star [1]

Any **two** from:

- Each device connects to a central node
- All communication goes through the central node
- Failing connection will only affect the device at the end of the connection
- Failure of the central node will affect the entire network
- Can easily add additional nodes

(max [2]) [3]

OR

Ring [1]

Any **two** from:

- Has a token which carries messages around nodes on network
- Node will only accept the message if it is addressed to it/if message addressed to another node it will be ignored
- Communication is in one direction only
- If one of the connected nodes fails the network will fail

(max [2]) [3]

(d) Any **two** from:

- Expensive peripherals can be shared
- Single copy of software is stored on file server and can be shared with users
- Communication between devices is possible
- Users can share files
- Users can log on to any computer to access their files [2]

(e) Allows devices to connect to the network/Provides a port for the network cable [1]
Allows devices to communicate with the server [1] [2]

(f) A router is a networking, hardware device or [1], in some cases, software in a computer, that determines the next network point to which a packet should be forwarded toward its destination or forwards data from one network to another. [1] [2] 16

10 (a)

Malware	Tick (✓)
Firewall	
Worm	✓
Router	
Virus	✓

[2]

(b) (i) Any **two** from:

- Data is encoded by sending computer
- Uses a software key
- Data can only be decoded if recipient has same software key [2]

(ii) Any **two** from:

- Prevents unauthorised access to the network
- Prevents viruses from entering the network
- Can prevent users from sending data out of the network [2] 6

AVAILABLE MARKS

AVAILABLE MARKS

11 (a) Any **two** from:

- Use of remote servers
- Via the internet
- To store/access/process data
- Instead of your own hard drive [2]

(b) Level 0 [0]
Answer is not worthy of credit.

Level 1 ([1]–[2])
The candidate refers to **one** [1], **two** [2] or **three** [2] advantages or describes one advantage of cloud computing to an organisation. The candidate makes only a limited selection of advantages and use of an appropriate form and style of writing. The organisation of the material may lack clarity and coherence. There is a little use of specialist vocabulary. Presentation, spelling, punctuation and grammar may be such that intended meaning is not clear.

Level 2 ([3]–[4])
The candidate describes **two** [3] or **three** [4] advantages of cloud computing. The candidate makes a reasonable selection of advantages and use of an appropriate form and style of writing. Relevant material is organised with some clarity and coherence. There is some satisfactory use of appropriate specialist vocabulary. Presentation, spelling, punctuation and grammar are sufficiently competent to make meaning evident.

Level 3 ([5]–[6])
The candidate fully describes all **three** advantages of cloud computing to an organisation. The candidate selects and uses an appropriate form and style of writing. Relevant material is organised with good degree of clarity and coherence. There is widespread use of appropriate specialist vocabulary. Presentation, spelling, punctuation and grammar are of a sufficiently high standard to make meaning clear.

Answers may include:

- Access to automatic software updates
- Cuts the cost on hardware needs
- Staff can access resources from anywhere
- Helps with collaboration/updates available in real-time
- Provides secure back-up/supports disaster recovery
- Increased security/files stored in secure environment/files not being sent as attachments
- Easy to increase storage capacity/cloud capacity if business expands

[6] 8

12 Level 0 ([0])
Answer is not worthy of credit.

AVAILABLE MARKS

Level 1 ([1]–[2])
The candidate mentions **one** [1], **two** [2] or **three** [2] of the areas identified or discusses **one** area. The candidate makes limited use of spelling, punctuation and grammar. The meaning of the text is not always clear. The candidate demonstrates a limited form and style appropriate to the question. The organisation or the answer is limited. There is little use of specialist terms.

Level 2 ([3]–[4])
The candidate discusses **two** of the areas. The candidate makes satisfactory use of spelling, punctuation and grammar. The meaning of the text is usually clear. The candidate demonstrates a satisfactory form and style appropriate to the question. The organisation of the answer is satisfactory. There is some use of specialist terms.

Level 3 ([5]–[6])
The candidate discusses all **three** areas. The candidate makes good use of spelling, punctuation and grammar. The meaning of the text is always clear. The candidate demonstrates good form and style appropriate to the question. The organisation of the answer is good. There is good use of specialist terms.

Answers may include:

Definition cyber space:
- Computer network of worldwide network of computers
- Virtual space where digitised information is communicated
- Use of Internet to store/modify/exchange data
- It describes the virtual space in which the electronic data of worldwide PCs circulate

Descriptions of any of the following two examples:
- Hacking
- Pornography
- Cyber stalking
- Data theft
- Denial of service
- Digital forgery
- Cyber defamation
- Spamming [6] 6

	AVAILABLE MARKS
13 (a) Any **three** from: • Provide regular eye tests • Plan work so employees can take regular breaks • Provide appropriate work environment/description of ergonomic furniture • Provide appropriate equipment/description of ergonomic hardware • Provide appropriate training in the use of technology [3]	
(b) Any **two** from: • RSI • Back strain • Eye strain [2]	5

14

Definition	Term
A radio navigation system that allows users determine their exact location	GPS [1]
Legislation implemented to give you rights when shopping online	Consumer contracts regulations [1]
A situation where a long-standing employee is removed from their original job and provided with an alternative job within the same organisation	Job Displacement [1]
A processing method where the input is dealt with quickly enough so as to be able to control an output properly	Real time processing [1]
A private network or website only available to members of an organisation. Not limited by geographic location	Intranet [1]
A wireless technology standard for exchanging data over short distances	Bluetooth [1]

[6] 6

Total **90**